RING HOMOMORPHISMS OUT OF A FIELD INTO A NONZERO RING ARE INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The reason F-algebra structures on nonzero rings are faithful: every morphism of extensions is an embedding.

Editorial note. The label lemma-field-maps-injective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Unital ring homomorphisms satisfy phi(1) = 1; a nonzero ring has 1 not equal to 0; nonzero elements of a field are invertible

Lemma 0.1. *If F is a field and R is a nonzero ring, then any ring homomorphism $\varphi : F \to R$ is injective.*

Proof. Indeed, let $a \in \mathrm{Ker}(\varphi)$ be a nonzero element. Then we have $\varphi(1) = \varphi(a^{-1}a) = \varphi(a^{-1})\varphi(a) = 0$. Thus $1 = \varphi(1) = 0$ and R is the zero ring. □

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